

Japji Soni

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EDUCATION

Chitkara University Institute of Engineering & Technology Rajpura, Punjab, India (2019-2023)

Btech | Major: Computer Science | Grade - 9.74 CGPA

WORK EXPERIENCE

Data Scientist - NLP at Birdeye, Gurugram, Haryana (June 2024 – present)

Birdeye is an online retail management SaaS platform based in Palo Alto, California, providing various types of chatbots and workflow agents under the new marketing automation program to its customers which include enterprise level accounts like DHL, Aspen Dental, Dave Hot Chicken, Hillary Beach Dental Clinic etc.

- Engineered production-grade voice AI agents for **Birdeye's Myna, an Operations AI Coworker serving healthcare and automotive verticals**. Owned end-to-end architecture for appointment booking, cancellation, and rescheduling flows, designing EHR-integrated identity verification pipelines, multi provider slot negotiation logic, and HIPAA compliant conversation guardrails. Built an AI-driven call quality recommendation system that analyzes live agent interactions, surfaces actionable coaching insights, and drives continuous improvement in call containment and resolution rates. Translated complex clinical workflows into natural, text-to-speech-optimized conversational experiences enabling autonomous, enterprise-grade patient interactions trusted by top dental and automotive clients.
- Currently developing an Email Template Copilot Agent within the **Marketing Automation** ecosystem, designed to help users create and execute multiple business specific campaigns. The agent leverages **LangGraph** based workflows and advanced copilot architectures to generate highly aesthetic, production-grade HTML email **templates from simple natural language prompts. Built as an end to end solution, it enables users to automate template creation**, personalize content at scale, and manage all marketing campaigns seamlessly within a single platform.
- Built an AI-driven content generation pipeline that incorporates **NLP** for automated post reasoning (strategic hook extraction) and provides contextual image recommendations using semantic embeddings and **AWS OpenSearch 2.9 k-NN FAISS** search.
- Contributed to **"Robin"** Birdeye's proprietary chatbot service, enhancing user experience for listed businesses by implementing RAGs (Retrieval-Augmented Generation) for an accurate knowledge base.
- Developed a Recommendation Generation Model for reviews data, capable of suggesting relevant improvements to businesses based on negative feedback snippets. This model achieved approximately 86% accuracy after fine-tuning **Meta-Llama-3-8B**.
- Created **Insights Summary Generation and Brand Voice** features, extracting performance insights from Birdeye-listed businesses and generating customized, brand-aligned summaries for owners to track their progress.

Associate Software Engineer at Createbytes, Gurugram, Haryana September 2023 – June 2024

1. Development of a Legal Compliance App:

- Engaged in creating a ML service which has revolutionized the legal compliance system of product packaging, providing an advanced and efficient compliance checking system. Our Service produced an **overall accuracy of about 95%.**
- I was involved in creating object detection modules using yolov8 ,prompt structuring on GPT 4 apis, regex for text extraction, image analysis on product images using opencv.
- **My approach of using langchains for prompt compression reduced the GPT-4 api costing by 28%.**

Skills Learnt: Azure Form Recognizer,Azure Service Bus Queues,Aws Textract,Fast Apis,Docker,Langchain memory buffers, fine tuning YOLOv8 model for object Detection,Regex,Threading in Python, Git.

2) Involved in creating a **user recommendation system** for an application named “Oter” which is a book recommendation app having 100K+ app downloads on play store. **My approach to use vector databases brought the query time from 5 mins to 30 seconds in order to filter similar posts for personalisation.**

Skills Learnt: Redis, Vector Databases - Pinecone and Chroma db.

3) Developed a **contextual search feature** for news feeds using **machine learning** to enhance search accuracy and efficiency beyond traditional keyword-based engines. Integrated sentence transformers for word-level embeddings to analyze news text contextually and identify relevant documentation. Integrated the machine learning service with the backend using **Django REST Framework** ,implemented secure login and signup functionality for users, ensuring authentication and personalized access to the platform. **Skills Learnt:**

Django Rest Framework , ORM, Serializers, Authentication,Model ViewSets,Websockets.

Technical Skills:

- **Programming & Scripting:** Proficient in **Python** with hands-on experience in building scalable AI applications.
- **Agentic AI & RAG Systems:** Experience designing and deploying agentic workflows and Retrieval-Augmented Generation (RAG) pipelines using vector databases and embedding models.
- **Cloud & Infrastructure:** **Worked with AWS Bedrock, Microsoft Azure (Service Bus & Queues), Firebase, and Kafka** for scalable and event-driven architectures.
- **Prompt Engineering & LLMs:** Expertise in crafting optimized prompts and fine-tuning large language models (LLMs) for domain-specific chatbots and reasoning tasks.
- **Machine Learning & MLOps:** Applied Deep Learning techniques and MLOps practices for model training, deployment, and monitoring.
- **Computer Vision & NLP:** Strong background in OpenCV and NLP, including custom pipelines and transformer-based architectures.

- Familiar with Networking Concepts such as TCP/IP , SSL handshake,IPv4 and IPV6.